Sylvester: A Benchmark and Method for Lifecycle-Aware Entity Resolution on Open Financial Data

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ABSTRACT

We introduce Sylvester, a public benchmark for *lifecycle-aware* entity resolution on financial entities: matching records across mergers, rebrands, spinoffs, and jurisdiction migrations where the surface form has deliberately changed. Built from public GLEIF Level-1/2, SEC EDGAR, and Wikidata, it provides 125,519 labeled pairs over 3.3M entities across three held-out evaluation regimes (time, event type, jurisdiction). Our central finding: a 22M-parameter MiniLM cross-encoder fine-tuned on this data reaches 0.82–0.87 AUROC across all three held-out regimes and 57–61 % Hits@1 on the time and event splits (±2.0–2.3 pp 95 % CI), 24–35 pp above the strongest bi-encoder across the three regimes and, under a uniform single-pair prompting protocol, above all six zero-shot LLMs tested (four organizations) on those splits. The lift concentrates on M&A, absorption, and breakup events, where the post-event surface form retains little of the predecessor's name.

Two caveats bound this result. The benchmark deliberately mines name-similar hard negatives against name-different positives, so the near-chance AUROC of most similarity baselines is a construction choice, not a claim about name matching in ER generally: an *inverted*-similarity baseline exploits it for 64–70 % Hits@1, and a supervised feature matcher — the production default — posts the best on-pool Hits@1 we observe (0.73–0.84). Both collapse once candidates come from an unrelated miner or from the full universe, where only the cross-encoder stays strong; that separation is what the benchmark is for. And in a fielded retrieve-then-rerank simulation, end-to-end Hits@1 falls to 15–28 % because bi-encoder Recall@20 caps at 33–50 % — the bottleneck is the first stage, not the reranker.

KEYWORDS

entity resolution, record linkage, financial entities, lifecycle events, GLEIF, benchmark, cross-encoder

1 INTRODUCTION

Twitter Inc. became X Corp. Facebook became Meta. Sprint was absorbed into T-Mobile. Pipelines silently breaking on such transitions is a familiar failure mode across fintech engineering, quant research, and compliance. The records look completely different on the surface — different legal names, different tickers, sometimes different jurisdictions and legal forms — yet they describe the same continuing economic entity.

Pair-matching ER — including LLM-based methods that achieve 99% F1 on point-in-time records [8] — treats each record as a snapshot of identity. Temporal entity-linking benchmarks such as TempEL [10] address mention-to-KB disambiguation against an evolving knowledge base, a different task. Financial KG construction [1] works at a single time slice. To the best of our knowledge no public benchmark targets the specific question we care about: *given two records observed at different times, do they describe the same economic entity, accounting for any lifecycle event that may have occurred between them?*

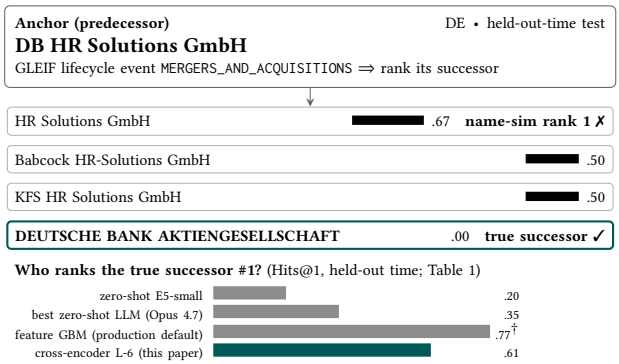


Figure 1: The task and the headline result. *Top:* a real anchor from the held-out-time test set (4 of its 7 candidates shown; bars = token Jaccard). GLEIF records a completed lifecycle event on the predecessor; methods must rank the true successor above deliberately name-similar hard negatives. Name similarity ranks a coincidental collision first and the true successor — zero shared tokens — last (§6.1). *Bottom:* on the released pool the supervised feature baseline (†) posts the best score by *exploiting* that construction; it collapses once the candidate pool shifts (Figure 3), while the cross-encoder holds.

A natural concern: the benchmark is engineered against name similarity by construction — negatives are mined name-similar while lifecycle successors are, by definition, name-different. This is deliberate, documented (§3.4, §5.4), and measured: we report the inverted-similarity exploit as its own baseline. The question the benchmark asks is the discriminative sub-problem a two-stage pipeline faces after a name-aware retriever narrows the field: *given a name-similar candidate pool, what architecture correctly identifies the true lifecycle successor?* (Figure 1.)

Contributions.

(1) **An open benchmark** (§3): 125,519 labeled pairs over 3.3M GLEIF Level-1 entities, hard negatives in four classes (the parent/subsidiary class drawn from GLEIF Level-2 relationship records), and three held-out regimes (time, event type, jurisdiction). Construction choices and their scope of generalizability are documented throughout (§3, §7).

(2) **A diagnostic finding** (§5): most similarity-based methods score near-chance AUROC on the time and event splits because negatives are mined to be name-similar while positives are name-different by construction—not a claim about ER generally. The construction cuts both ways: *inverted* name similarity and a supervised feature matcher exploit it for the best on-pool scores we observe, yet both collapse under pool shift (§5.4). A 22M-parameter cross-encoder is the only method strong across every condition we test (24–35 pp Hits@1 over the best bi-encoder), with gains concentrated on M&A and absorption events and an advantage that survives a mechanically different miner (§5.6).
(3) **The deployment bottleneck is retrieval, not reranking** (§5.6): in a fielded simulation over 3.3M entities, end-to-end Hits@1 falls to 15–28 % because bi-encoder Recall@20 caps at 33–50 %. The cross-encoder reranks effectively (46–57 % conditional Hits@1); the system breaks at the first stage, not the second.

An enriched-prompt L-6×L-12 ensemble lifts Hits@1 a further 9–11 pp on time/event (Table 2); baseline comparisons use the single fine-tuned L-6 cross-encoder.

2 RELATED WORK

Classical ER [4] matches records via similarity over structured fields; modern PLM cross-encoders on serialized records (Ditto [7], GraLMatch [3]) are now standard, and the OpenSanctions Pairs corpus [8] benchmarks LLM matching at 755K pairs — all under a *snapshot* framing, not lifecycle-spanning matches. TempEL [10] time-stratifies entity linking against an evolving Wikipedia, but that task is mention-to-KB disambiguation, not record-to-record matching over legal entities. Financial-filing knowledge graphs (FinReflectKG [1], from this conference) are constructed at a single time slice; the lifecycle-bridge question is orthogonal.

3 THE SYLVESTER BENCHMARK

3.1 Data sources

We construct Sylvester entirely from open, persistently-licensed sources:

- **GLEIF Level-1 LEI-CDF.** The LEI repository (3.3M entities, 2026-05-19 snapshot), CC0, refreshed every eight hours. Each `LEIRecord` carries legal name, jurisdiction, legal form, addresses, registration metadata, a dated `LegalEntityEvents` log, and a `SuccessorEntity` field linking predecessor to successor by LEI.
- **GLEIF Level-2 Relationship Records.** 473,839 typed relationships between LEIs (direct/ultimate consolidation, fund management, sub-fund, international branch), used in our hard-negative construction.
- **Wikidata.** High-profile M&A via SPARQL on the P156 (*followed by*) property, restricted to businesses; a marquee supplement and cross-validation source.
- **SEC EDGAR.** `company_tickers.json` (10,354 records) gives a name↔CIK crosswalk to GLEIF for the U.S. public subset (3,334 confident links); released with the benchmark, not used in the evaluation.

3.2 Positive labels

Our gold-standard positives are GLEIF `SuccessorEntity` links: explicit predecessor-LEI → successor-LEI pairs maintained by GLEIF's network of Local Operating Units. The 2026-05-19 snapshot contains 50,202 records with a non-null `SuccessorEntity` element; 33,645 of these have a resolvable `SuccessorLEI` (the rest reference successors that never registered an LEI and are excluded).

Label quality validation. A five-class auto-classifier on the successor pairs (using documented event types, normalized name delta, jurisdiction, legal form) yields: `strong_real` (documented event + name change) 25,428; `likely_real_name_changed` 3,330; `likely_real_event_only` 1,704; `ambiguous` 1,355; **`likely_administrative`** 1,828 (5.4%, filtered out). We retain the first four classes (94.6%; `ambiguous` stays because every link is still LOU-curated), yielding 31,817 plausibly-real predecessors (31,824 positive pairs; some predecessors have multiple successors).

3.3 Hard negatives

For each positive predecessor we mine up to 80 hard negatives drawn from four classes:

- `parent_subsidiary` (221 pairs) — LEIs that share a Level-2 relationship-record edge (consolidation, fund-management, sub-fund) with the predecessor. These look related but are legally distinct.
- `same_name_same_country` (10,208) — distinct LEIs with identical normalized legal names in the same jurisdiction: coincidental name collision vs. true lifecycle linkage.
- `same_name_diff_country` (10,703) — distinct LEIs with identical normalized name in a different country.
- `token_overlap_same_country` (232,601) — fuzzy fallback: distinct LEIs in the same country sharing ≥ 2 content tokens after normalization, ranked by Jaccard. Covers the long tail when name-exact collisions are sparse.

After deduplication: 253,733 candidate (predecessor, hard-negative) pairs covering 20,138 of 31,817 predecessors (~63%). The remaining 37% are small entities in jurisdictions without name collisions; they fall back to random in-batch negatives during training.

3.4 Evaluation splits

For each positive we identify its *anchor lifecycle event* — the highest-priority COMPLETED event on the predecessor's record by type-priority then effective date. We then assign each positive (and its associated negatives) to one of three split regimes:

- **Held-out time:** anchor effective date < 2025-01-01 (train) vs. ≥ 2025-01-01 (test). Completed-event volume rises ~5× from 2022, when GLEIF tightened event reporting (distribution figure in the released repo), so the effective window is 2022–2026. Test composition: 3,448 positives; 2,007 anchors are Hits@1-eligible (§5.2).

- **Held-out event type:** train on M&A, rename, legal-form change; test on spinoff, demerger, breakup, absorption. The test set is heavily dominated by *absorption* (2,609 of 2,760 anchors, 94.5%); demerger (120), breakup (25), and spinoff (6) are present but statistically thin; per-event-type metrics within the split are reported in §5.
- **Held-out jurisdiction:** train on US + GB + DE; test on SG + JP + BR. Test composition: 161 anchors, 161 positives, of which 88 are Hits@1-eligible — the smallest split; its headline numbers carry ∼±8.5 pp 95% bootstrap CIs (vs. ∼±2.0–2.3 pp on time/event).

Each positive gets up to 5 hard negatives if its splits are all `train`, or up to 10 if any split places it in `test`, yielding 125,519 final pair rows.

Wikidata marquee supplement. Released alongside the benchmark: 302 high-profile M&A pairs from Wikidata P156 both-sides-resolved to GLEIF, 283 not captured by GLEIF `SuccessorEntity` (Sprint → T-Mobile, Wyeth → Pfizer, Interbrew → AB-InBev); not used in the headline evaluation.

4 METHOD

We present two methods at different cost-accuracy points: a bi-encoder (§4.1) that embeds each side independently and scores by cosine — cheap, but hemmed in by the benchmark's anti-correlation property (name similarity anti-predicts the label here; §5.4) — and a cross-encoder (§4.2) that reads both records together, 4–10× more expensive per pair, producing the headline result.

4.1 Bi-encoder

We fine-tune `intfloat/e5-small-v2` (33M params, 384-dim) via supervised contrastive learning with state features folded into the text prompt: "`query:` n_ℓ `|` `jurisdiction:` c_ℓ `|` `form:` f_ℓ". Training uses the 26,297 positives in the strict no-leakage intersection of all three split regimes (one hard neg per anchor), MultipleNegativesRanking loss (InfoNCE; in-batch + explicit hard neg as negatives), 3 epochs, batch 32, AdamW lr= 2×10^{-5}, 100 warmup, ∼6 min on Apple-silicon (MPS). Ablation variants in Table 1: `method_no_hard` (in-batch negs only), `method_v2_full` (adds E5-cosine hard negs + 1,372 transitive positives). Loss is contrastive; the cross-encoder below uses BCE on the logit — different objectives by architectural convention, not strictly controlled.

4.2 Cross-encoder

A bi-encoder must place each record at a fixed point in embedding space, yet the event generating each positive deliberately changes the record's surface form — it is asked to keep cosine-close two records that look very different on the page. A cross-encoder reads both records jointly and can compare specific tokens between them.

We fine-tune a second-stage cross-encoder that ingests $(\text{record}_A, \text{record}_B)$ as a single concatenated sequence and emits a single relevance logit. The backbone is the 22M-parameter `ms-marco-MiniLM-L-6-v2` cross-encoder from HuggingFace; training uses the same positives as `method_v2_full` (26,297 plus 1,372 transitives) plus a 4:1 sample of E5-cosine hard negatives per positive, with positives labeled 1.0 and negatives 0.0. Loss: standard binary cross-entropy on the logit. 2 epochs, batch 32, lr 2×10^{-5}, 100 warmup steps. The extensions in §5.3 hold architecture/optimizer constant and modify only the serialized text, the training positives, and the backbone.

Inference cost. Scoring all 125,519 pairs in the benchmark takes ≈80 seconds on MPS. In deployment the cross-encoder would re-rank a retriever's top-K (the standard two-stage IR pattern); the benchmark's fixed 5–10-negative pools make direct full scoring feasible, and Table 1 reflects that setting.

Hard-negative re-mining. Re-mining with E5-cosine top-20 same-country candidates, in place of the noisy token-overlap fallback, yields 652,764 negatives across 31,714 predecessors (vs. 253,733 over ∼20K before). The cleaner negatives are what the cross-encoder trains on, to large effect; for the bi-encoder they change nothing (§5.7).

5 EXPERIMENTS

5.1 Baselines

We compare against four similarity baselines — `exact_name` (normalized name equality), `token_set` (token Jaccard), `tfidf_char` (char-3gram TF-IDF cosine), and `sentence_encoder` (zero-shot E5-small-v2, the method's backbone un-tuned) — and one *supervised* classical matcher, `features_gbm`: gradient-boosted trees over twelve string-similarity and field-agreement features in the Fellegi–Sunter lineage [4], trained on the released pool's no-leakage rows; the production default. Six zero-shot LLMs run under one protocol: one call per pair, byte-identical prompts, a 0–10 integer rubric, `temperature=0` where exposed. Five API models from three organizations cover the full test set (26,564 pairs each) — **Claude Haiku 4.5**, **Sonnet 4.6**, **Opus 4.7**, **Gemini 3.1 Pro**, **DeepSeek V4 Pro** (thinking disabled) — and **gpt-oss-20b** (a fourth organization) runs locally on a 56/52/50-anchor sample (Table 1 footnotes).

Paired permutation tests on per-anchor expected Hits@1 separate the comparisons cleanly. The single L-6 cross-encoder beats all five API models on TIME and EVENT at $p < 10^{-4}$ (+15 to +37 pp). On the 88-anchor JURIS split it leads Haiku, Gemini, and DeepSeek ($p \leq 0.018$), is within noise of Sonnet ($p = 0.14$), and is statistically indistinguishable from Opus 4.7, which holds the higher point estimate (0.767 vs. 0.750, $p = 0.74$). The L-6×L-12 ensemble beats Opus on TIME/EVENT ($p < 10^{-4}$); JURIS is a coin flip (0.784 vs. 0.767). Figure 2 shows the panel: zero-shot quality is strikingly vendor-heterogeneous — every Claude model outscores both other API vendors' frontier models on every split, including Haiku, the cheapest model tested; §7 examines the elicitation-format confound behind part of this spread.

5.2 Metrics

We report per-anchor retrieval metrics (Hits@1, MRR) computed by ranking each anchor's candidates by score, and AUROC for the pair-wise classification framing; Hits@3, AUPRC, and threshold-tuned F1 ship with the released per-run metrics. Discrete scorers tie frequently within a pool, so Hits@k/MRR are exact *expectations*

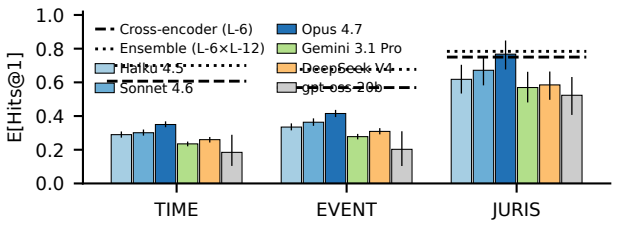


Figure 2: Zero-shot LLM panel (uniform single-pair protocol) vs. fine-tuned rerankers (lines): E[Hits@1], 95% CIs. Every LLM sits far below the 22M cross-encoder on the two large splits; on JURIS (n=88) Opus exceeds the L-6 point estimate (n.s.). Haiku outscores both other API vendors' frontier models everywhere.

over uniformly random tie-breaking (a fixed tie order is biased). Per-anchor metrics cover *eligible* anchors — pool size ≥ 2 with the gold present. Because 37% of predecessors have no mined hard negative (§3), their singleton pools are excluded: e.g., 1,612 of 2,760 EVENT-test anchors are eligible.

5.3 Headline results

Table 1 reports test-set metrics across the three split regimes for all baselines and method variants on the released benchmark pool (`pairs.parquet`; token-overlap-mined negatives). Headline cross-encoder Hits@1 95% bootstrap confidence intervals are ±2.0 pp (TIME), ±2.3 pp (EVENT), and ±8.5 pp (JURIS, $n = 88$ anchors); the jurisdiction-split gain comfortably clears its CI band but absolute numbers on that split should be treated as directional. We retest the headline cross-encoder on a mechanically different candidate pool (§5.6) and find the gap shrinks but does not collapse; Table 2 extends the L-6 cross-encoder to the L-6×L-12 ensemble.

Extensions: enriched prompt + L-6×L-12 ensemble. Stacking four modifications on the headline cross-encoder — prompt enrichment (GLEIF Level-2 parent/child names, founding year, ISO country expansion, category), Wikidata-LEI identifier mining (259 training positives via P156/P1366/P7888; distinct from §3's marquee supplement), multi-seed averaging, and ensembling with an L-12-backbone variant — lifts Hits@1 a further +9 to +11 pp on time/event (+3 on jurisdiction, inside its CI band; Table 2). The ensemble reaches 0.700 / 0.677 / 0.784 Hits@1; the headline result reported against baselines remains the single L-6 cross-encoder, so that comparison stays apples-to-apples. A negative result is worth recording: symmetric scoring (averaging $s(a, b)$ and $s(b, a)$) hurts −0.9/−1.9/−9.1 pp, so the cross-encoder really does learn a directional predecessor→successor representation.

Backbone size does not break the bi-encoder ceiling. Scaling E5-small (33M) to E5-base (110M) leaves TIME/EVENT Hits@1 unchanged (−0.2/−1.2 pp) and lifts JURIS +2.3 pp: the constraint is structural, not parameter-bound — cosine-close embeddings for lexically-different pairs are not a problem more parameters fix.

5.4 The construction cuts both ways: an anti-similarity baseline

`token_set`, `tfidf_char`, and `sentence_encoder` all land below 0.5 AUROC on the held-out-time and held-out-event-type splits (Table 1). The cause is mechanical: hard negatives are deliberately name-similar to the predecessor while positives are by construction name-different, so on this candidate distribution name similarity is *evidence against* a true successor link. So we report the mirror image as its own baseline: `anti-token_set` ranks candidates by *negated* Jaccard, and on the curated pool it is strong (Table 1; *above* the single L-6 cross-encoder on time and event) because picking the least name-similar candidate exploits the construction.

The supervised `features_gbm` baseline is the exploit's learned apotheosis: given training labels, it discovers the inversion plus the miners' residual signatures, and posts the best on-pool numbers in this paper (Table 1) — above the L-6×L-12 ensemble on TIME/EVENT ($p < 10^{-4}$ paired permutation; JURIS n.s.). Off-pool it collapses (Figure 3; JURIS: 0.36 on Sylvester-TF-IDF, 0.12 fielded; AUROC 0.54–0.57 and ≈0.5 respectively). Retraining the GBM on the harder TF-IDF negatives buys real but partial robustness (Figure 3, third row; JURIS: 0.69 cross-pool, 0.31 fielded): negative *quality* matters for feature models, and on the TF-IDF pool itself this variant *outscores* the cross-encoder (0.59 / 0.60 vs. 0.54 / 0.53) — a feature model trained on a pool's own miner is hard to beat on that pool. What it does not buy is transfer: it drops to 0.53 on the released pool and 0.31–0.34 fielded, trailing the cross-encoder's fielded conditional by 17–20 pp. Our claim is therefore about the *minimum* across conditions, not a win in every cell: the cross-encoder is the only method here that never falls below 0.52, and it never needed a matched miner to get there.

The unsupervised exploit shows the same pattern at lower amplitude, surviving every pool change we test but one event type. On Sylvester-TF-IDF it collapses to chance (0.25 / 0.25 / 0.21 Hits@1) while the cross-encoders hold 0.53–0.66 (Table 4); on the fielded pool it drops to 9–18 % conditional Hits@1, a fraction of the cross-encoder's (§5.6). Per event it is a one-way bet on acquirer-renaming, and there it is genuinely hard to beat: on the M&A slice of the TIME split (n=1,506) it scores 0.741, *above* both the L-6 (0.625) and the ensemble (0.711). That is the one condition we test where an exploit is not merely competitive but best, and it is the event type the inversion is built for. The bet is one-way: the same heuristic scores 0.11 on pure renames and 0.35 on legal-form changes, where the cross-encoder scores 0.90 and 0.85, and it loses pairwise AUROC to the cross-encoder on every split (Table 1). Curated-pool Hits@1 alone can therefore be gamed; the cross-encoder's claim is being strong *simultaneously* on the curated pool, the alternative-miner pool, the fielded pool, and every event type with a usable sample (Figure 3). The mechanism is joint attention: asking whether the *differences* between two records look like a lifecycle-event pattern, not how similar two strings are.

5.5 Per-event-type breakdown

Aggregate Hits@1 hides large per-event-type variation.

Held-out time. Pure renames (`CHANGE_LEGAL_NAME`, $n = 10$) are essentially solved by every similarity-direction method including

Table 1: Test-set metrics on the three split regimes, released pool. Bold = headline L-6 cross-encoder used in baseline comparisons; the L-12 matches or beats it throughout, and the L-6×L-12 ensemble (Table 2) is the best method that survives pool shift (§5.4). Random ranking scores 0.18–0.22 Hits@1 (pools average 7.4–8.3 candidates); the best bi-encoder lifts Hits@1 5–8 pp over zero-shot E5-small and the cross-encoder a further 24–35 pp. Hits@1 95% CIs: ±2.0 / ±2.3 / ±8.5 pp. Both exploit rows — [§]the supervised GBM over string+state features, and `anti-token_set`, which ranks by *negated* similarity — are strongest here and fall to near-chance AUROC on every other pool (§5.4, Table 4). Protocol and metrics: §5. [†]Opus: no temperature param; Gemini (preview): thinking not disableable, lowest level, Batch API. [‡]local; 56/52/50-anchor sample.

	Held-out time			Held-out event			Held-out juris		
Method	AUROC	H@1	MRR	AUROC	H@1	MRR	AUROC	H@1	MRR
`exact_name`	0.474	0.163	0.373	0.523	0.193	0.400	0.571	0.352	0.539
`token_set`	0.193	0.136	0.319	0.266	0.180	0.356	0.323	0.356	0.522
`tfidf_char`	0.277	0.180	0.360	0.349	0.213	0.386	0.568	0.420	0.583
`sentence_encoder`	0.315	0.204	0.385	0.386	0.232	0.408	0.648	0.432	0.587
`anti-token_set` (1−Jaccard)	0.807	0.700	0.761	0.734	0.638	0.715	0.677	0.516	0.618
`features_gbm` (supervised)[§]	0.935	0.772	0.840	0.926	0.734	0.816	0.960	0.835	0.897
LLM pairwise, zero-shot — uniform single-pair protocol, full test coverage									
`Claude Haiku 4.5`	—	0.290	0.476	—	0.335	0.517	—	0.618	0.738
`Claude Sonnet 4.6`	—	0.301	0.492	—	0.364	0.542	—	0.672	0.758
`Claude Opus 4.7`[†]	—	0.350	0.534	—	0.416	0.589	—	0.767	0.839
`Gemini 3.1 Pro`[†]	—	0.235	0.433	—	0.278	0.474	—	0.569	0.699
`DeepSeek V4 Pro`	—	0.260	0.451	—	0.309	0.497	—	0.585	0.712
`gpt-oss-20b`[‡]	—	0.185	0.377	—	0.203	0.403	—	0.523	0.683
`method_full` (bi-enc, token-overlap negs)	0.403	0.255	0.425	0.451	0.291	0.450	0.667	0.500	0.627
`method_no_hard` (bi-enc, no explicit negs)	0.407	0.259	0.429	0.461	0.299	0.459	0.668	0.511	0.641
`method_v2_full` (bi-enc, E5-cosine negs)	0.405	0.250	0.424	0.457	0.296	0.455	0.694	0.500	0.629
Cross-encoder (L-6, 22M)	**0.836**	**0.607**	**0.725**	**0.825**	**0.569**	**0.701**	**0.868**	**0.750**	**0.818**
Cross-encoder (L-12, 33M)	0.844	0.631	0.742	0.836	0.598	0.724	0.870	0.750	0.823

Table 2: Cross-encoder extensions: Hits@1 [95% bootstrap CI]. Headline L-6 row repeated from Table 1. Prompt enrichment is the largest single increment on TIME/EVENT; the L-12 pairing captures the JURIS gain (its CIs overlap across the chain — read per-step deltas as directional).

Model	TIME	EVENT	JURIS
Cross-encoder (L-6, 22M)	.607 [.587,.627]	.569 [.546,.592]	.750 [.665,.835]
+ enriched prompt	.675 [.653,.697]	.641 [.620,.662]	.750 [.659,.830]
+ Wikidata-LEI positives	.694 [.669,.712]	.651 [.630,.673]	.773 [.682,.853]
+ L-12 backbone (33M)	.697 [.678,.716]	.676 [.657,.697]	.761 [.670,.841]
Ensemble L-6×L-12	**.700 [.681,.718]**	**.677 [.656,.700]**	**.784 [.682,.864]**

Table 3: Per-event-type Hits@1 on the held-out-EVENT test split (Hits@1-eligible subset), zero-shot E5-small vs. L-6. The full test split is 94.5% `ABSORPTION` (2,609 of 2,760); `SPINOFF` at $n = 4$ is statistically uninformative. Lift concentrates in the absorption-like types.

Event type	*n* anchors	zero-shot E5	Cross-enc (L-6)	Δ
`ABSORPTION`	1,546	0.230	**0.571**	+34.1 pp
`DEMERGER`	42	0.333	**0.595**	+26.2 pp
`BREAKUP`	20	0.150	**0.400**	+25.0 pp
`SPINOFF`	4	0.500	0.250	−25.0 pp[†]

[†] $n = 4$, statistically uninformative.

zero-shot E5-small (0.90 Hits@1, no lift); only the inverted exploit fails them (0.11, §5.4). The headline gains come from the events where the post-event name comes from the acquirer and string similarity cannot bridge the gap: `MERGERS_AND_ACQUISITIONS` (n = 1,506: 0.18 zero-shot → 0.62 L-6 → 0.71 ensemble; the inverted exploit reaches 0.74 here) and `ABSORPTION` (n = 461: 0.24 → 0.54 → 0.65). The ensemble also more than doubles zero-shot on the small but operationally critical `BREAKUP` subset (n = 7: 0.14 → 0.71).

"Held-out event type" really means "held-out absorption." The held-out-event test set is 94.5% `ABSORPTION` anchors (§3.4); `DEMERGER`, `BREAKUP`, and `SPINOFF` contribute $n = 42$, $n = 20$, and $n = 4$ within the Hits@1-eligible subset. Table 3 shows the lift is consistent across the absorption-like types and that the `SPINOFF` cell is statistically uninformative, so the aggregate "+34 pp Hits@1 over zero-shot E5" on this split is primarily a transfer claim about absorption, not a strong claim about spinoffs.

5.6 Robustness to candidate-pool shift

The headline cross-encoder result is trained and (mostly) evaluated against candidates from the same family of hard-negative miners — v1 token-overlap and v2 E5-cosine both rank candidates by name-related signals, and the cross-encoder trains on E5-cosine negs. The obvious question: does the gap survive an evaluation pool built with a mechanically different miner, or is the headline number partly an artifact of how the negatives were chosen?

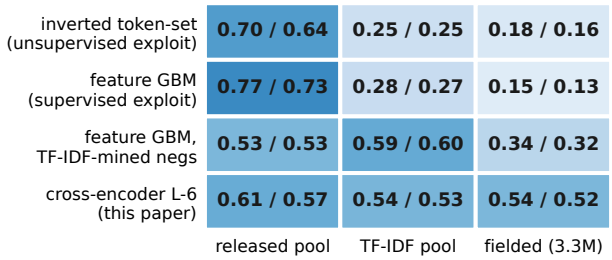


Figure 3: The argument in one exhibit: expected Hits@1 (TIME / EVENT; fill = mean) as the candidate pool shifts from the released pool to an unrelated miner's pool to retrieval from the full 3.3M-entity universe. Both exploits collapse off-pool. Retraining the GBM on TF-IDF-mined negatives wins the pool matching its own miner (0.59 / 0.60, middle column) but does not transfer; read rows by their *minimum*, where only the cross-encoder stays above 0.52. The TF-IDF-pool cross-encoder cell is the re-trained `trn-E5` reproduction (Table 4), which matches the headline model within 2 pp on TIME/EVENT. Fielded cells include the gold for evaluability — strict retrieval-conditional and end-to-end numbers in the text; JURIS (n=88) in Tables 1 and 4.

Setup. We re-mine hard negatives with sparse char-3gram TF-IDF cosine (same signal as the `tfidf_char` baseline) — chosen because it has no learned representation and shares no parameters with E5. The other three classes (parent/subsidiary, same-name same/diff-country) are held constant. We call the resulting pair set Sylvester-TF-IDF; splits and positives are unchanged. We train two cross-encoders: one on the E5-cosine hard negatives (trn-E5), one on the TF-IDF negatives (trn-TF-IDF), and score each on both pair pools. Note the released pool's negatives are token-overlap-mined, so the headline configuration (E5-trained, benchmark-pool evaluation) is itself already cross-miner; the only miner-matched cell in the matrix is trn-TF-IDF on the TF-IDF pool.

What the 2×2 tells us. First, the cross-encoder *architecture* generalizes across miners: the TF-IDF-trained model evaluated on the benchmark pool lands within 1–2 pp of the E5-trained model on TIME/EVENT (4.5 pp ahead on thin JURIS, inside its CI) despite never seeing that pool's miner, and the E5-trained model on the TF-IDF pool stays well above every non-cross-encoder method. Second, in-distribution alignment is real but bounded: pool shift costs the E5-trained model 2–13 pp, and matching the training miner to the evaluation pool is worth +7–11 pp — non-trivial, but not the headline gap. Third, the gain over zero-shot E5 is robust to the miner: +20 to +40 pp in the headline pairings and +13 to +39 pp swapped, well outside any CI band. Joint encoding does real work on lifecycle ER — but the absolute size of the gain depends on how the candidate pool was mined, so the "+27–35 pp" figure in the abstract should be read in that light.

Fielded retrieve-then-rerank. The miner-robustness experiment above shifts the mining signal but still draws candidates from a curated negative pool. A real deployment pipeline does something different: a first-stage retriever pulls the top-K candidates from the full entity universe, then a second-stage model reranks. We simulate that here. The trained `method_v2_full` bi-encoder retrieves top-20 candidates per test anchor from the full 3.3M-entity GLEIF universe, restricted to the anchor's jurisdiction (the lightest filter a production system would apply); the cross-encoder reranks. The filter is nearly free (1.8–4.1 % of test positives cross countries, capping recall at ≥96 %, far above the observed 33–50 %): the bottleneck is the embedding, not the filter.

Bi-encoder Recall@20 is 33.2 / 39.6 / 49.7% on TIME / EVENT / JURIS. That is the hard ceiling for any reranker. When the bi-encoder does retrieve the true successor, the L-6 cross-encoder picks it #1 with 46–57% Hits@1 and the L-6×L-12 ensemble reaches 53–60% — the ensemble's edge over the L-6 is modest once distractors look genuinely lifecycle-plausible. End-to-end deployment Hits@1 (recall × conditional, un-retrieved truths counted as misses) is 15–28% for the L-6 cross-encoder and 18–30% for the ensemble. The drop from 57–61% in-distribution to 15–30% deployment confirms rather than contradicts §4.2: the bi-encoder's fixed-embedding limit, which fails to retrieve lifecycle-spanning successors by cosine similarity, is exactly the bottleneck the anti-correlation analysis predicts.

5.7 Ablations

State features help modestly. Ablating jurisdiction and legal form from the prompt costs 1.5–2 pp Hits@1 on every split; the country token contributes most on held-out-jurisdiction, least on held-out-event-type.

Explicit hard negatives slightly *hurt* the bi-encoder (Table 1): `method_no_hard` edges out the v1 full model by 0.4–1.1 pp Hits@1 on every split, and E5-cosine re-mining does not change that. The culprit is the `token_overlap_same_country` class, 92% of mined v1 negatives and the lowest-precision one; in-batch negatives from MultipleNegativesRanking are cleaner on their own. So better explicit negatives do not always help contrastive training — but they are exactly what the cross-encoder needs (§4.2).

5.8 Additional diagnostics

Four diagnostics; full tables ship with the released code. **B-cubed cluster F1** [2] on the test partition is 0.92 / 0.90 / 0.91 for the cross-encoder vs. ≤0.47 for any bi-encoder (each of which collapses to one giant cluster at the tuned threshold). Restricting to records in true clusters of size ≥ 2 — 64% of test records are singletons — gives a more honest 0.85 / 0.84 / 0.88. **Calibration and selective prediction.** ECE (10-bin) 0.20, Brier 0.18 — over-confident in the high-score bin, and a train-fitted temperature does not fix it, so we report uncalibrated scores. A selective-prediction [5] sweep on the ensemble reaches Hits@1 ≥ 0.94 at 20% coverage and 0.87–0.97 at 50%, so high-precision users can threshold. **Cross-domain transfer.** Zero-shot Magellan Walmart-Amazon [6]: the cross-encoder *loses* to its own zero-shot base encoder by 5.9 pp AUROC. The fine-tuned representation is specialized to lifecycle-spanning legal-entity matching, not generic ER. **Label quality, second opinion.** A rule-based re-classifier — orthogonal to the primary's documented-event-type evidence, though still name-informed (name-stem similarity, cross-script checks, legal-form

Table 4: Hits@1 with 95% bootstrap CIs for the 2×2 train/evaluation matrix. "Benchmark pool" = `pairs.parquet` (the released pool; token-overlap-mined negatives dominate); "TF-IDF pool" = `pairs_v3.parquet` (TF-IDF char-3gram cosine, no neural representation). Row labels indicate which miner's hard negatives were used for cross-encoder training. The E5-small zero-shot reference row is bootstrapped separately and differs from Table 1 by ≤0.002. The cross-encoder advantage is large and consistent in the headline pairings (+20 to +40 pp) and persists when pairings are swapped (+13 to +39 pp). Re-trained `trn-E5` cells reproduce the original L-6 predictions within 2 pp on time/event.

	Eval on benchmark pool (`pairs.parquet`)			Eval on TF-IDF pool (`pairs_v3.parquet`)		
	TIME	EVENT	JURIS	TIME	EVENT	JURIS
E5-small zero-shot	0.203	0.234	0.432	0.265	0.319	0.441
	±1.8	±2.1	±10.5	±1.4	±1.7	±7.8
cross-enc trn-E5	**0.600**	**0.551**	0.705	0.545	0.528	0.571
	±2.2	±2.4	±8.8	±1.8	±1.7	±7.5
cross-enc trn-TF-IDF	0.588	0.533	**0.750**	**0.655**	**0.614**	**0.640**
	±2.0	±2.4	±9.1	±1.5	±1.7	±7.1
Δ *vs. E5, headline pairing*	+39.7	+31.7	+27.3	+39.0	+29.5	+19.9
Δ *vs. E5, swapped pairing*	+38.5	+29.9	+31.8	+28.0	+20.9	+13.0

continuity) — agrees with the primary auto-classifier on 98.7% [95% CI 97.0–100%] of the 231/277 stratified-sampled positives it judges confidently; the remaining 46 (16.6%) are marked uncertain. This bounds heuristically detectable label noise below 2% there; human expert review is future work.

5.9 Where each method is fooled

Table 5 reports per-negative-class mean score (higher = more often fooled). The fine-tuned bi-encoders cut the zero-shot E5 baseline's score on `parent_subsidiary` and `same_name_diff_country` negatives by 40–52%, but `same_name_same_country` remains a wall at ≈0.89 mean score: the bi-encoder cannot reliably tell a real successor link from two distinct entities sharing an exact normalized name in the same country.

The cross-encoder breaks that wall: `same_name_same_country` mean score drops from 0.89 (best bi-encoder) to 0.43, and the L-6×L-12 ensemble drops it further to 0.36. The ensemble also nearly solves `same_name_diff_country` (0.91 zero-shot → 0.17) and `token_overlap_same_country` (0.91 → 0.28). The one regression is `parent_subsidiary` (0.50 bi-encoder → 0.67 L-6 cross-encoder → 0.75 ensemble): the joint encoder occasionally treats parent–subsidiary edges as positive lifecycle bridges, and the ensemble's prompt enrichment that injects GLEIF L2 parent/child names makes the regression *worse*, not better. With only 99 held-out examples in this class the headline impact is small, but the trend points at a clear direction: explicit multi-class modeling of corporate-hierarchy relationships.

Table 5: Negative-class mean score (lower is better) on negatives *never seen in training* (test in ≥1 regime; n = 99 / 1,398 / 1,096 / 20,759), so supervised and zero-shot rows are scored out-of-sample alike. *p/s* = parent/subsidiary, *sn-D* = same-name/diff-country, *sn-S* = same-name/same-country, *tok* = token-overlap same-country. The ensemble improves three classes but *worsens p/s* (0.67 → 0.75) — prompt-injected L2 context makes parent/subsidiary pairs look more similar, not less. Scores are on each method's own scale; compare down columns, not across rows; bold = lowest among the learned methods.

Method	p/s	sn-D	sn-S	tok
`exact_name`	0.010	1.000	1.000	0.000
`token_set`	0.132	1.000	1.000	0.488
`tfidf_char`	0.220	0.590	0.783	0.595
`sentence_encoder`	0.841	0.912	0.960	0.905
`method_full`	0.501	0.438	0.893	0.747
Cross-encoder (L-6, 22M)	0.668	0.201	0.432	0.311
Ensemble (L-6×L-12)	0.748	**0.166**	**0.358**	**0.284**

6 ANALYSIS

6.1 Error analysis

We sampled 100 `method_full` failures on the held-out-time test split (positives not ranked first) and categorized them via simple text features: 66 mixed/other, 19 where anchor and true successor share *no* tokens, 8 with substring overlap with a wrong candidate, and 7 exact normalized-name collisions with a wrong candidate.

The 19 "no shared tokens" cases are the hardest: no similarity-based representation can solve them — they require world knowledge or graph context (board overlap, regulatory filings, press releases). Figure 1's anchor is representative: resolving "DB" to Deutsche Bank is external knowledge a 33M-parameter encoder is unlikely to memorize from name strings alone.

Residual JURIS failures cluster on sibling consolidation. Of the 19 / 88 anchors the ensemble mis-ranks on JURIS, ≥11 are sibling consolidations (e.g., 9 "`EPIC <port> PTE. LTD.`" Singapore ships into `BW EPIC KOSAN VESSELS ONE`; the model returns sibling `EPIC` vessels); a second cluster is Japanese-script name changes the ASCII tokenizer cannot bridge (Daiwa-Sumigin → Mitsui-Sumitomo DS). Multilingual pretraining plus ownership-graph context would address both — though §5 shows zero-shot prompting alone is not the answer.

6.2 Why this matters in practice

The benchmark targets a failure mode common across financial infrastructure: transaction and merchant embeddings at scale [9], vendor and invoice deduplication, and the entity backbones under market-data and compliance stacks all face the lifecycle-ER gap after a merger, rebrand, or absorption. Sylvester lets practitioners test whether their system handles it.

7 LIMITATIONS AND FUTURE WORK

The "anti-correlation" finding is a construction artifact. Near-chance AUROC for similarity baselines follows directly from mining name-similar negatives against name-different positives (§5.4); it motivates joint encoding here, not a universal claim about name similarity in ER.

In-distribution headline overstates deployment. The headline Hits@1 is measured on the released pool; end-to-end on the fielded pipeline it falls to 15–30 % (§5.6) because bi-encoder Recall@20 is 33–50 %, so the reranker often never sees the true successor. The gain is partly pool-specific but survives miner shift (+13 to +39 pp; Table 4). A stronger first stage (hybrid sparse–dense, or a pre-filtered shortlist) is the highest-leverage future work.

Seed variance and a sliver of mislabeled hard negatives. Over seeds 42/7/1337 the bi-encoder varies ±0.5/0.4/1.6 pp and the enriched cross-encoder ±0.5/0.3/1.6 pp (TIME/EVENT/JURIS) — JURIS's spread tracks its small sample. Separately, 205/93,695 = 0.22% of hard negatives appear in GLEIF's `SuccessorEntity`; dropping them moves results ≤0.2 pp.

Survivorship bias and thin slices. GLEIF `SuccessorEntity` is voluntarily populated, biasing positives toward well-counseled regulated entities; of 302 marquee Wikidata M&A pairs, 283 are absent from it (§3.4), so high-profile events are under-represented. The held-out-event test set is 94.5% absorption and the jurisdiction test has 88 eligible anchors (±8.5 pp CI): gains clear the CI band on all three splits, but per-split absolutes should be read with those sample sizes in mind. Eligibility itself conditions on the predecessor having a name-colliding neighbour, which amplifies the construction (§5.2). A 277-pair second-classifier check (§5.8) bounds detectable label noise at <2% on the confident subset. Splits are pair-level, not entity-disjoint (32 % / 21 % of time-split test candidates / gold successors appear in training pairs; JURIS <1 %), though L-6 Hits@1 shows no seen-entity advantage (unseen 0.62 vs. seen 0.57 TIME; 0.56 vs. 0.58 EVENT, both n.s.); entity-disjoint splits are a stricter future design.

LLM baselines: what the discrete scale hides. A 0–10 rubric caps what ranking can express: models forfeit 13–48 pp Hits@1 to within-pool ties, and Gemini 3.1 Pro scores every candidate identically on 55% of TIME anchors — oracle tie-breaking would give it 0.72 vs. 0.23 measured, above our L-6 — while Sonnet/Opus collapse only ∼5% of pools and the cross-encoder's continuous scores pay no tie tax. Prompt format moves results as much (batched Sonnet 0.43 TIME vs. 0.30 single-pair). Our panel therefore bounds what this elicitation format extracts, not what these models can do: finer scales, log-probs, or comparative tie-breaks would likely raise them, and cross-vendor gaps may partly reflect exposure to GLEIF-derived text.

The model is specialized, and parent/subsidiary is a thin slice. The cross-encoder underperforms zero-shot E5-small on out-of-domain product matching (Walmart-Amazon, −5.9 pp AUROC; §5.8): the representation is specialized to lifecycle ER, not generic. Parent/subsidiary negatives regress (0.50 → 0.67 → 0.75 across bi-encoder, L-6, and ensemble), though at 99 held-out examples the effect is weak; explicit multi-class modeling of related-but-distinct entities is clear future work.

8 CONCLUSION

Sylvester is an open benchmark for lifecycle-aware entity resolution on financial entities, built entirely from public GLEIF, SEC EDGAR, and Wikidata data. Its central result is deliberately two-sided. On the released pool the construction can be gamed: inverted name similarity and a supervised feature matcher post the best on-pool scores in the paper. Shift the candidate pool — an unrelated miner, or retrieval from the full 3.3M-entity universe — and those exploits collapse, while a 22M-parameter cross-encoder is the only method whose *worst* condition stays strong; the fielded pipeline is then bottlenecked by first-stage retrieval rather than reranking. Each exploit still wins the one slice built to favour it (§5.4), which is the point: the benchmark's value is its discriminating power, separating methods that learn the pool from methods that learn the task.

Reproducibility. Code, parsers, predictions, trained weights, and a datasheet: https://github.com/chang-07/er-research. Every non-LLM result reproduces from raw GLEIF in ∼2 hours on one MacBook.

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